Filipino Adaptation of OmniVoice for Voice Cloning and Text-to-Speech: A Pilot Study

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Abstract

This pilot study asks whether Filipino-specific fine-tuning improves OmniVoice, a pretrained massively multilingual zero-shot text-to-speech and voice cloning model. OmniVoice already supports Filipino, but its language table reports only 7.71 hours of Filipino in the original multilingual training mixture. We prepare a cleaned 39.8-hour Filipino read-speech corpus from the UP-DSP Philippine Languages Database, tokenize the audio with the Higgs Audio tokenizer, fine-tune the pretrained checkpoint on free Kaggle T4 GPUs, and evaluate generated speech on a 4,322-utterance speaker-disjoint test set using word error rate (WER), objective speaker similarity (SIM-o), and UTMOS. Four systems are compared in a controlled setup: the base model and the best-development-loss checkpoints of three 5000-step fine-tuning runs that differ only in learning rate (5e-6, 1e-5, 2e-5). The 1e-5 checkpoint, selected at step 4900, is the strongest overall: it reduces WER from 22.55% to 18.52% (a 17.9% relative reduction) while keeping SIM-o slightly above the base model. The 2e-5 checkpoint nearly matches that WER at 18.83% but drops SIM-o to 0.583, below the base model, while the 5e-6 checkpoint preserves speaker similarity best (SIM-o 0.605) but improves WER by only 0.59 points. The base model keeps the highest UTMOS. Filipino fine-tuning can therefore improve intelligibility under limited compute, but the learning rate controls an intelligibility-versus-speaker-similarity tradeoff, and checkpoints cannot be judged from training loss or any single metric alone.

CCS Concepts

• **Computing methodologies → Natural language processing**; *Neural networks.*

Keywords

text-to-speech, voice cloning, Filipino, low-resource languages, transfer learning, OmniVoice, Philippine Languages Database

1 Introduction

Voice cloning and text-to-speech (TTS) systems synthesize speech from written text and, in zero-shot voice cloning settings, imitate the voice of a reference speaker from a short audio prompt. Recent neural speech generation systems rely on pretrained multilingual models, discrete audio tokenizers, and generative modeling strategies to support many languages and speakers without training a new model for every use case [8, 13–15]. These systems matter for accessibility, education, narration, localization, and language-learning applications.

Filipino is a major language in the Philippines, but open Filipino voice cloning workflows remain less developed than English-centered systems. Training a Filipino model from scratch would require data and compute far beyond a course-scale project. A more practical path is transfer learning [10]: start from an existing multilingual TTS model and fine-tune it on a curated Filipino dataset.

OmniVoice is a suitable base model for this approach because it is open source, supports more than 600 languages, and includes Filipino with language ID `fil` [15]. Broad coverage, however, does not guarantee strong language-specific quality. OmniVoice reports 7.71 hours of Filipino in its multilingual training mixture, while the UP-DSP Philippine Languages Database (PLD) holds 48:56:36 of Filipino speech [4]. The gap this study addresses is therefore not missing Filipino support; it is that Filipino-specific performance may remain limited because Filipino is a small fraction of the original training data, and a curated corpus with roughly four times more usable Filipino training speech is available to close that gap.

The machine learning problem is: given a pretrained multilingual voice cloning model and a curated Filipino speech-text corpus, determine whether supervised fine-tuning improves Filipino generated speech compared with the original pretrained checkpoint. The main research question follows directly: how much does Filipino-specific fine-tuning improve OmniVoice for Filipino TTS and voice cloning across intelligibility, naturalness, and speaker-similarity metrics?

To answer it, the study makes the following contributions:

(1) a cleaned, speaker-disjoint Filipino read-speech dataset (42,276 utterances, 39.8 hours) prepared from PLD in the JSONL and tokenized WebDataset format OmniVoice expects;
(2) a reproducible fine-tuning workflow for the pretrained `k2-fsa/OmniVoice` checkpoint that fits free Kaggle T4 GPU limits, with all runs tracked in Weights and Biases [2];
(3) a controlled comparison of the base model and the best-development-loss checkpoints of three identical 5000-step fine-tuning runs differing only in learning rate, evaluated on the full held-out test split with WER, SIM-o, and UTMOS;
(4) an error and trade-off analysis explaining which fine-tuning configuration helps, which hurts, and why checkpoint selection matters.

The study is limited to Filipino read speech from PLD. It does not train a model from scratch, evaluate other Philippine languages, or claim commercial readiness. Evaluation is objective and automatic; human listening evaluation is a recommended next step. Kaggle's free T4 environment constrains model size, training steps, batch size, and checkpoint retention throughout.

2 Related Work

Voice cloning is a specialized form of TTS where the system must generate the requested content while preserving speaker identity from a reference prompt [1]. Recent work has shifted from speaker-specific systems toward large pretrained speech generation models. VALL-E demonstrated neural codec language modeling for zero-shot TTS by generating discrete audio codec tokens conditioned on text and a short acoustic prompt [13]. Voicebox extended multilingual speech generation and text-guided speech editing at scale [8]. MaskGCT and F5-TTS reflect a broader move toward efficient non-autoregressive or masked generation [6, 14].

OmniVoice is the model most directly connected to this study. It is a massively multilingual zero-shot TTS system with a diffusion language model-style masked-token prediction process [15]. Instead of predicting semantic tokens first and then generating acoustic tokens, OmniVoice directly maps text and conditioning signals into multi-codebook acoustic tokens. It uses the Higgs Audio tokenizer for 8-codebook acoustic representation and waveform reconstruction [3]. These design choices make it practical to fine-tune an existing pretrained checkpoint rather than build a new Filipino speech generator from scratch.

The UP-DSP Philippine Languages Database is the primary data source. Cajote et al. [4] describe PLD as a multilingual speech corpus for Philippine spoken languages, covering Filipino, English, Cebuano, Kapampangan, Hiligaynon, Ilokano, Bikolano, Waray, and Tausug. The Filipino subset provides the paired speech and transcript files that supervised TTS fine-tuning requires.

Evaluation in zero-shot TTS usually separates intelligibility, speaker preservation, and naturalness. WER or CER measures intelligibility through an ASR model [9, 11], speaker-embedding cosine similarity measures objective speaker preservation [5, 7], and UTMOS estimates naturalness [12]. OmniVoice and Voicebox report these same metric families [8, 15]; this study adapts that evaluation logic to Filipino fine-tuning.

3 Methodology

Figure 1 shows the system architecture across data preparation, tokenization, Kaggle fine-tuning, evaluation, and public delivery.

3.1 Dataset and Pre-processing

The dataset is the Filipino subset of the UP-DSP Philippine Languages Database, released through Mozilla Data Collective under a CC-BY-NC-4.0 license. It consists of WAV audio files and LOG transcript files. The PLD paper reports 135 speakers, 52,879 utterances, 48:56:36 total duration, and a 3.58-second average utterance for the Filipino subset [4].

Only read or prompted Filipino speech was used. Spontaneous speech was excluded because a speaker's free response may not match the displayed prompt, which raises transcript-audio mismatch risk for supervised TTS. The preparation script parses PLD folders, matches WAV files to LOG transcript entries, filters unsuitable rows, and writes OmniVoice-compatible JSONL manifests. Each row contains an utterance ID, relative audio path, transcript text, and language ID:

```
{"id":"0000.110816.021250.0001",
 "audio_path":"wavs/0000/....0001.wav",
 "text":"manugang",
 "language_id":"fil"}
```

The filtering rules were intentionally conservative: spontaneous prompt sources were dropped, transcripts containing parentheses or digits were excluded, unreadable or zero-byte WAV files were removed, only clips between 1.0 and 15.0 seconds were kept, and transcript whitespace was lightly normalized. Speakers were split disjointly across train, development, and test sets so that no test voice appears in training. Table 1 shows the resulting splits.

Table 1: Prepared Filipino dataset splits (speaker-disjoint).

Split	Samples	Percent	Hours	Speakers
Train	33,448	79.12%	31.90	103
Development	4,506	10.66%	3.71	13
Test	4,322	10.22%	4.17	13
Total	42,276	100.00%	39.77	129

The cleaned corpus spans varied prompt domains. The largest read-speech sources are news sentences (4,330 clips), medical sentences (2,397), time expressions (1,973), common expressions (1,941), short stories (1,904), greetings (1,708), interrogatives (1,634), and Filipino proverbs or *salawikain* (1,620). Many remaining sources are isolated-word prompts (names, cuisines, landmarks, ordinals), which explains the short average utterance length and matters later when interpreting prosody-related results.

The 31.9 training hours are roughly four times the 7.71 Filipino hours in OmniVoice's original training mixture, which is the quantitative basis for expecting fine-tuning to help.

3.2 Audio Tokenization

OmniVoice does not train on raw waveforms. Audio is resampled to 24 kHz and converted into discrete 8-codebook acoustic token sequences with the Higgs Audio V2 tokenizer[1] [3], then stored in WebDataset shards referenced through `data.lst` manifests. Tokenization is expensive on free GPUs, so the tokenized full dataset was saved once as a Kaggle dataset artifact and reused across all training runs. Text needs no feature engineering beyond keeping the Filipino transcript and the `fil` language ID.

3.3 Fine-Tuning Setup

All runs fine-tune the pretrained `k2-fsa/OmniVoice` checkpoint, whose language model component is `Qwen/Qwen3-0.6B`, inside Kaggle T4 notebooks. Weights and Biases tracked configurations, losses, checkpoints, generated examples, and evaluation summaries

[1] `eustlb/higgs-audio-v2-tokenizer` on the Hugging Face Hub.

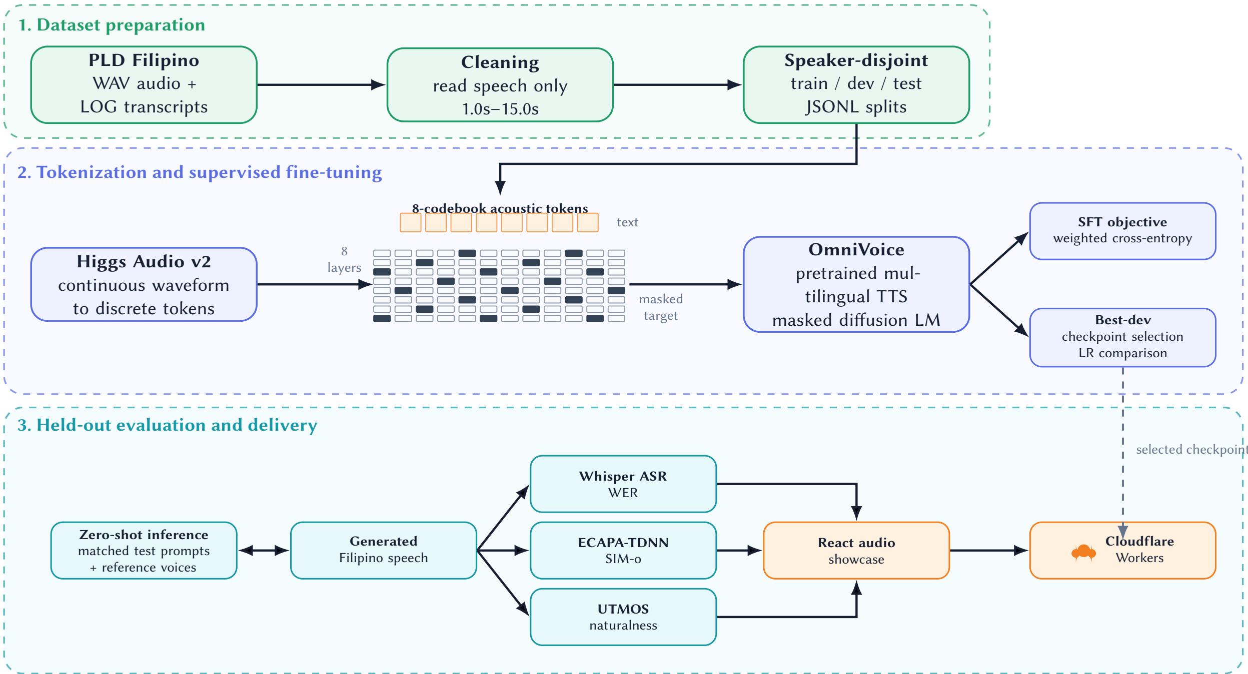

Figure 1: System architecture for Filipino OmniVoice fine-tuning, evaluation, and public showcase delivery.

Table 2: Shared training configuration across all fine-tuning runs.

Field	Value
Base model	`k2-fsa/OmniVoice`
LLM path	`Qwen/Qwen3-0.6B`
Audio tokenizer	Higgs Audio V2
Audio codebooks / vocab	8 / 1025 (mask ID 1024)
Precision / attention	FP16 / SDPA
Batch tokens	1024
Gradient accumulation	8
Max batch size	4
Sample tokens (min–max)	50–1200
Optimizer	AdamW, weight decay 0.01
Max gradient norm	1.0
LR schedule	cosine decay, 1% warmup
Loss	weighted cross-entropy on masked acoustic tokens
Codebook loss weights	[8, 8, 6, 6, 4, 4, 2, 2]
Mask ratio range	[0.0, 1.0]
Prompt ratio range	[0.0, 0.3]
Drop condition ratio	0.1
Seed	42

[2]; the consolidated report is available online.[2] Table 2 lists the shared configuration.

[2]https://wandb.ai/duncanb013-polytechnic-university-of-the-philippines/OmniVoice-PLD-Filipino/reports/OmniVoice-PLD-Filipino-finetuning--VmlldzoxNjk0ODkyMg?accessToken=ktcztpc3ocu4gmtctis13mdubb82ec1r4rnyuxdta56tts4z5j4f965olbm0v5ka

The loss is OmniVoice's own training objective. OmniVoice is a single-stage non-autoregressive model trained with a discrete masked diffusion objective [15]: random target audio-token positions are replaced by the mask token, and the model recovers the original token IDs only at those positions, conditioned on text, language, prompt audio, and unmasked target tokens. Our runs implement this as weighted cross-entropy over masked acoustic-token positions; text tokens, control tokens, prompt audio, and unmasked targets are excluded from the loss. The codebook weights [8, 8, 6, 6, 4, 4, 2, 2] emphasize lower codebooks, which carry coarser speech structure.

The comparison uses three controlled runs that share the same 5000-step budget and the same development-loss-selected checkpoint policy — the development set is evaluated at fixed intervals and a checkpoint is saved only when development loss improves — and differ only in learning rate:

- **FT-5e-6:** the most conservative update; its lowest development loss occurred at the final step, 5000.
- **FT-1e-5:** the repository-aligned fine-tuning learning rate; its lowest development loss, 4.162, occurred at step 4900.
- **FT-2e-5:** the most aggressive update; its lowest development loss occurred at the final step, 5000.

The learning rates are deliberately lower than OmniVoice's large-scale pretraining setting because these runs update an already pretrained checkpoint on a much smaller corpus. The 1% warmup gives 50 warmup steps per run.

Table 3: Full PLD Filipino test-set results. All fine-tuned rows are best-development-loss checkpoints of identical 5000-step runs differing only in learning rate. Lower WER is better; higher SIM-o and UTMOS are better. Bold marks the best value per column.

Model	WER (%)	ΔWER	SIM-o	UTMOS
Base OmniVoice	22.55	—	0.602	**3.64**
FT-5e-6 (step 5000)	21.96	−0.59	**0.605**	3.60
FT-1e-5 (step 4900)	**18.52**	−4.03	0.604	3.61
FT-2e-5 (step 5000)	18.83	−3.72	0.583	3.61

3.4 Evaluation Design

All systems were evaluated on the same held-out 4,322-utterance test split with identical manifests, inference parameters, and evaluator models, so differences between rows reflect the checkpoints rather than the pipeline.

For each test utterance, the model generates the target text while cloning a reference voice. The reference prompt is a *different* utterance from the same test speaker (the next utterance from that speaker in the manifest), so the model never hears the target audio it must produce. Inference used 32 unmasking steps with batch size 2 on two T4 GPUs. The base model was measured in a first full evaluation run; later runs evaluated each best-development-loss fine-tuned checkpoint and reused the base row, which is valid because the test split, manifests, inference parameters, and evaluator versions were unchanged.

Three objective metrics were computed with the OmniVoice evaluation modules and the `k2-fsa/TTS_eval_models` evaluator checkpoints:

- **WER** (lower is better): generated audio is transcribed by Whisper-large-v3 [11] with Tagalog as the decoding language, and the transcript is compared with the target text by edit distance [9]:

$$\mathrm{WER} = \frac{S + D + I}{N} \times 100, \tag{1}$$

  where S, D, I are substitutions, deletions, and insertions and N is the number of reference words. WER is an ASR-mediated intelligibility estimate, not a direct human comprehension score.
- **SIM-o** (higher is better): cosine similarity between speaker embeddings of the reference prompt and the generated audio,

$$\text{SIM-o}(r, g) = \frac{\mathbf{e}_r \cdot \mathbf{e}_g}{\|\mathbf{e}_r\|\|\mathbf{e}_g\|}, \tag{2}$$

  using a WavLM-large ECAPA-TDNN speaker encoder [5, 7].
- **UTMOS** (higher is better): a learned mean-opinion-score predictor for naturalness [12], averaged over all generated test files.

4 Results and Discussion

4.1 Objective Metric Comparison

Table 3 and Figure 2 summarize the full-test-set comparison.

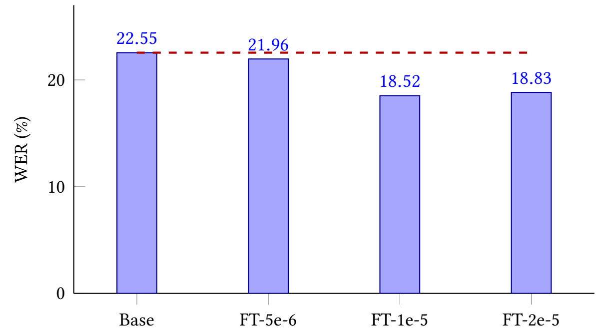

Figure 2: Test-set WER by system. The dashed line marks the base-model WER; all fine-tuned checkpoints fall below it, with FT-1e-5 and FT-2e-5 clearly ahead.

Table 4: WER edit-operation counts on the full test set.

Model	Ins.	Del.	Sub.	Words
Base OmniVoice	2,668	640	2,858	27,342
FT-5e-6	3,005	577	2,426	27,354
FT-1e-5	2,045	603	2,417	27,354
FT-2e-5	2,156	604	2,392	27,354

The base model obtained 22.55% WER, 0.602 SIM-o, and 3.64 UTMOS, confirming that OmniVoice is already a usable Filipino system before adaptation. Because the fine-tuned rows share the same step budget and checkpoint policy, the differences between them are attributable mostly to learning rate, and they form a consistent pattern.

The strongest result is FT-1e-5, selected at step 4900. It reached 18.52% WER, 4.03 absolute points below the base model and a 17.9% relative reduction, while keeping SIM-o slightly above base (0.604 versus 0.602). FT-2e-5 nearly matched that intelligibility at 18.83% WER, but it has the lowest SIM-o of all four systems at 0.583 — the only fine-tuned value below the base model — so its extra adaptation came at the cost of the reference voice. FT-5e-6 moved least: it preserved speaker similarity best among the fine-tunes (0.605) but improved WER by only 0.59 points.

UTMOS separates the systems least: the base model keeps the highest predicted naturalness at 3.64 and the fine-tuned checkpoints sit close together at 3.60–3.61. These SIM-o and UTMOS differences are small and should not be over-read, but their direction is consistent. The comparison is an intelligibility-versus-speaker-similarity tradeoff governed by learning rate, not a single-metric win; FT-1e-5 is the strongest overall checkpoint because it captures nearly all of the available intelligibility gain without giving up speaker similarity or much predicted naturalness.

4.2 Error Analysis

Table 4 and Figure 3 break WER into edit operations.

The edit counts explain the ranking. Compared with the base model, FT-1e-5 reduced insertions from 2,668 to 2,045 and substitutions from 2,858 to 2,417, while deletions stayed near the base level; its improvement comes from producing fewer extra words

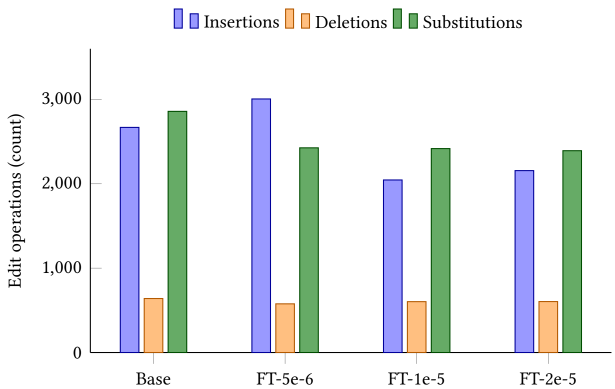

Figure 3: Edit operations per system. FT-1e-5 and FT-2e-5 win mainly by cutting insertions and substitutions; FT-5e-6 loses most of its substitution gain to an insertion increase.

and fewer wrongly recognized words. FT-2e-5 follows the same pattern with slightly more insertions and slightly fewer substitutions. FT-5e-6 instead cut substitutions to 2,426 but raised insertions to 3,005, above the base model, and that single increase absorbed most of its substitution gain. A fine-tuned model can therefore improve one error type while worsening another, which is why aggregate WER must be read together with its components.

4.3 Development Loss versus Downstream Metrics

Development loss was used to monitor training and select each run's checkpoint, and the relationship to downstream quality turned out to be useful but incomplete. Within-run, selecting the lowest-development-loss checkpoint gave every run a principled stopping point, and the selected FT-1e-5 checkpoint produced the lowest WER of the study. Across runs, however, development loss could not have predicted the full metric pattern: FT-1e-5 and FT-2e-5 are separated by only 0.31 WER points yet by 0.021 SIM-o, and FT-5e-6's insertion problem is invisible in its loss curve. Development loss is a good checkpoint-selection heuristic; checkpoints from different runs still must be compared with downstream audio metrics before declaring a winner.

4.4 Practical Findings

Several workflow-level findings emerged. Reusing tokenized audio as a Kaggle dataset artifact was essential because token extraction is expensive and would otherwise be repeated every run. The conservative T4 recipe (FP16, SDPA attention, small per-forward token batches, gradient accumulation) made training feasible at all, progressing at roughly 1000 steps per hour. Full-test evaluation is slow, so reusing the base-model row across evaluation notebooks is practical when the evaluation setup is frozen. Finally, the strongest model should be chosen by a coherent metric pattern rather than a single number: WER is primary here because intelligibility is the research question, while SIM-o and UTMOS guard against selecting a clearer model that damages voice similarity or naturalness.

4.5 Limitations

The completed comparison uses automatic metrics only. WER depends on the ASR model (Whisper-large-v3 decoding Tagalog), so it estimates intelligibility as that ASR system hears it, and ASR for Filipino is itself imperfect; SIM-o depends on the speaker encoder; and UTMOS is a learned MOS predictor, not a listening survey, so the small naturalness differences reported here are predictions rather than human judgments. No human listener study has been conducted. The corpus is Filipino read speech with many short, isolated-word prompts, so the results say little about spontaneous speech, long-form prosody, or other Philippine languages. Compute limits also restricted the number of hyperparameter runs, so the controlled learning-rate comparison samples three points rather than a full sweep, with one run per learning rate and no variance estimate across seeds.

5 Conclusion and Recommendations

Filipino-specific fine-tuning can improve OmniVoice for Filipino TTS and voice cloning, and in the controlled comparison the learning rate determines what kind of improvement is obtained. FT-1e-5 (best development loss at step 4900) reduced WER from 22.55% to 18.52% while keeping SIM-o slightly above the base model and is the recommended checkpoint. FT-2e-5 bought nearly the same intelligibility at a visible cost in speaker similarity, FT-5e-6 stayed closest to the base model on every axis, and the base model retained the highest predicted naturalness. The answer to the research question is therefore a tradeoff, not a uniform win: fine-tuning reliably improves intelligibility at well-chosen learning rates, while speaker similarity and predicted naturalness move within narrow margins whose direction depends on the learning rate.

Three directions follow. First, a Filipino listener study should validate whether the WER gain matches human judgments of pronunciation, intelligibility, naturalness, and speaker similarity; the automatic metrics used here are proxies for all four. Second, the training side should sample more learning rates around 1e-5, repeat runs across seeds, and inspect generated audio where WER remains high. Third, the data side should build a verified subset of longer sentence-level utterances, since the current corpus is dominated by short prompts, and test whether sentence-level training data improves Filipino prosody. For demonstration and reporting, the recommended checkpoint is FT-1e-5.

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A Supplementary Figures

The full interactive experiment report is available in the project's W&B workspace (footnote in Section 3.3). Figure 4 shows the final metric comparison across all three evaluation metrics, Figure 5 shows the W&B record of the recommended learning-rate 1e-5 run, and Figures 6 and 7 show the W&B generated-test-sample tables for the base model and the recommended fine-tuned checkpoint.

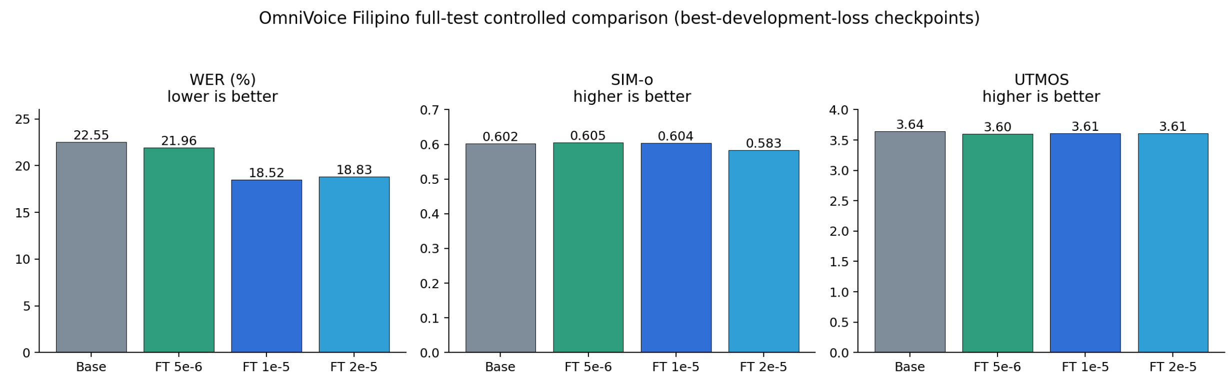
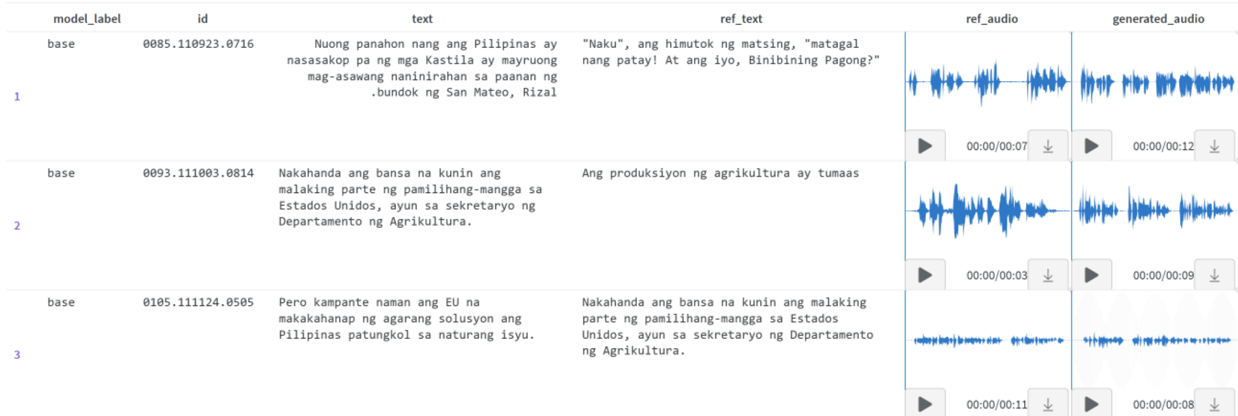


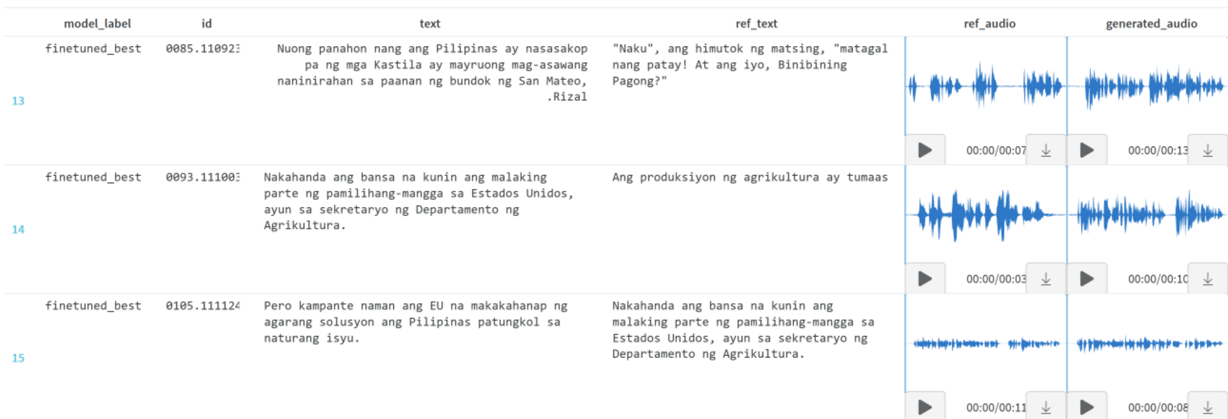
Figure 4: WER, SIM-o, and UTMOS for the final controlled comparison on the full PLD Filipino test split.

Figure 5: W&B run screenshot for the 5000-step fine-tuning run at learning rate 1e-5, whose best-development-loss checkpoint (step 4900) is the recommended model.



	model_label	id	text	ref_text	ref_audio	generated_audio
1	base	0085.110923.0716	Nuong panahon nang ang Pilipinas ay nasasakop pa ng mga Kastila ay mayruong mag-asawang naninirahan sa paanan ng .bundok ng San Mateo, Rizal	"Naku", ang himutok ng matsing, "matagal nang patay! At ang iyo, Binibining Pagong?"	00:00/00:07	00:00/00:12
2	base	0093.111003.0814	Nakahanda ang bansa na kunin ang malaking parte ng pamilihang-mangga sa Estados Unidos, ayun sa sekretaryo ng Departamento ng Agrikultura.	Ang produksiyon ng agrikultura ay tumaas	00:00/00:03	00:00/00:09
3	base	0105.111124.0505	Pero kampante naman ang EU na makakahanap ng agarang solusyon ang Pilipinas patungkol sa naturang isyu.	Nakahanda ang bansa na kunin ang malaking parte ng pamilihang-mangga sa Estados Unidos, ayun sa sekretaryo ng Departamento ng Agrikultura.	00:00/00:11	00:00/00:08

Figure 6: W&B generated-test-samples table for the base OmniVoice model.



	model_label	id	text	ref_text	ref_audio	generated_audio
13	finetuned_best	0085.11092	Nuong panahon nang ang Pilipinas ay nasasakop pa ng mga Kastila ay mayruong mag-asawang naninirahan sa paanan ng bundok ng San Mateo, .Rizal	"Naku", ang himutok ng matsing, "matagal nang patay! At ang iyo, Binibining Pagong?"	00:00/00:07	00:00/00:13
14	finetuned_best	0093.11100	Nakahanda ang bansa na kunin ang malaking parte ng pamilihang-mangga sa Estados Unidos, ayun sa sekretaryo ng Departamento ng Agrikultura.	Ang produksiyon ng agrikultura ay tumaas	00:00/00:03	00:00/00:10
15	finetuned_best	0105.11112	Pero kampante naman ang EU na makakahanap ng agarang solusyon ang Pilipinas patungkol sa naturang isyu.	Nakahanda ang bansa na kunin ang malaking parte ng pamilihang-mangga sa Estados Unidos, ayun sa sekretaryo ng Departamento ng Agrikultura.	00:00/00:11	00:00/00:08

Figure 7: W&B generated-test-samples table for the 5000-step learning-rate 1e-5 fine-tuning run.